AGLDM: ATTRIBUTE-GUIDED ZERO-SHOT TEXT-TO-IMAGE SYNTHESIS USING DATA-EFFICIENT LATENT DIFFUSION MODEL WITH SELF-CONSISTENCY LOSS

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ABSTRACT

Synthesizing images of objects not seen during training remains a major challenge in computer vision. In this paper, we introduce a method for synthesizing images of objects not seen during training from text prompts (zero-shot text-to-image synthesis), utilizing various attributes of objects as auxiliary information. This is motivated by the fact that a superior correlation between the text input and object attributes may facilitate superior image synthesis. We synthesize high-fidelity images using a latent diffusion model, requiring only thousands of training samples compared to the billions of samples required by existing large-scale models. To that end, we use salient visual features of images extracted by a vector-quantized variational autoencoder as input to the diffusion model for training. To ensure the quality of the synthetic images, we introduce a novel classifier that produces attributes from images. An attribute-based self-consistency loss for the classifier helps in improving the synthetic images. Quantitative and qualitative evaluations demonstrate the effectiveness of the proposed method. Our results also highlight the complementary nature of textual and attribute-based representations for an efficient zero-shot image synthesis. This opens a new avenue for generative models, especially in problems with limited visual examples.

***Index Terms*—** Zero-shot learning, text-to-image synthesis, latent diffusion, attribute guidance

1. INTRODUCTION

Generative models have achieved significant progress in synthesizing high-quality, diverse images. However, generating images of objects absent from the training set (known as zero-shot image synthesis) remains a challenging task. Such synthetic images have numerous applications, including rare species identification, medical anomaly synthesis, and industrial defect visualization.

Traditional generative approaches, including Generative Adversarial Networks (GANs) [1] and Variational Autoencoders (VAEs) [2], typically struggle with zero-shot image generation. This limitation stems from their fundamental dependence on learning the underlying data distribution directly from visual examples, making it difficult to generalize to novel categories unseen during training.

Recent text-to-image diffusion models like DALL-E [3] and Stable Diffusion [4] have shown promising results by leveraging semantic knowledge embedded in large language models. However, these approaches require massive computational resources and billions of training samples. Zero-shot learning (ZSL) addresses this limitation by incorporating auxiliary information about different objects. This auxiliary information, such as handcrafted attributes or structured representations, provides interpretable and structured visual characteristics of various objects [5].

In this paper, we propose AGLDM, a novel framework for attribute-guided zero-shot text-to-image synthesis using a latent diffusion model with self-consistency loss. Our method leverages object attributes as auxiliary information through a data-efficient latent diffusion architecture that requires only thousands of training samples compared to the billions required by existing large-scale models. The proposed design is motivated by several key insights.

First, textual descriptions provide rich semantic information about visual appearance in natural language, capturing subtle nuances. Second, auxiliary information in the form of attributes offers complementary benefits through explicit encoding of visual properties, enabling more precise control over specific features and providing structured cues about both seen and unseen objects during training. By training on seen classes (categories with training data) and leveraging both textual descriptions and attribute vectors, we generate high-fidelity images of unseen classes (categories completely absent from training). Figure 1 shows synthetic images generated by our model.

Third, establishing meaningful correlations between salient visual characteristics and semantic cues from text and attributes facilitates superior image synthesis. To achieve this, we utilize a vector-quantized variational autoencoder for extracting salient features from input images during training. Fourth, we employ a latent diffusion model that provides an efficient avenue to correlate visual and semantic features and

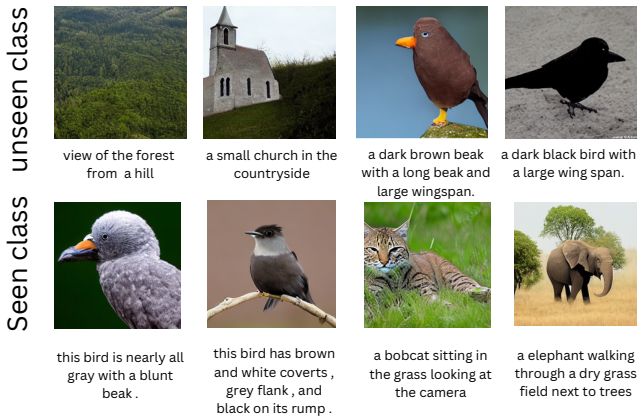

Fig. 1. Synthetic images of seen and unseen classes generated by AGLDM using text prompts and attributes, trained on only thousands of samples.

synthesize semantically meaningful images from text using attributes. Finally, we employ a classifier trained to predict attribute vectors and introduce a self-consistency loss. Minimizing this loss ensures that synthetic images are semantically grounded, facilitating the synthesis of high-fidelity, semantically meaningful images from unseen classes.

Our main contributions are: (1) We demonstrate that competitive zero-shot text-to-image synthesis can be achieved using only thousands of training samples on a single GPU, making the technology accessible to the broader research community. (2) We develop attribute guidance for traditional text-to-image diffusion processes, effectively combining rich semantic information from natural language descriptions with structured knowledge from attribute vectors. (3) We introduce a latent diffusion architecture specifically designed for zero-shot generation tasks, leveraging VQ-VAE to learn discrete visual concepts that can be shared across seen and unseen classes. (4) We introduce a novel self-consistency training objective that enforces attribute fidelity in generated images, creating a feedback loop that significantly improves semantic alignment.

2. METHODS

We propose a data-efficient zero-shot text-to-image generation model that addresses limitations of existing resource-intensive approaches. While textual descriptions provide rich semantic context, they often lack structured representation of specific visual features necessary for accurate zero-shot generation. Our key insight is that correlating attributes with image features in a shared latent space facilitates more effective knowledge transfer from seen to unseen classes using significantly fewer training samples.

Our approach consists of a two-stage pipeline as illustrated in Figure 2:

1. We first train a Vector Quantized Variational Autoencoder (VQ-VAE) [6] on images from seen classes (Phase One in Figure 2). The VQ-VAE learns a discrete codebook of visual features, representing images in a structured latent representation that facilitates efficient knowledge transfer across categories.

2. We then train a conditional latent diffusion model on representations from the VQ-VAE (Phase Two in Figure 2). This diffusion model learns to gradually denoise random latent vectors into coherent latent representations conditioned on text and guided by attribute information.

2.1. Problem Formulation

We define training samples as $\mathcal{S} = \{(x, y, a, t) | y \in \mathcal{Y}_S\}$, where x denotes an image, y its class label from seen classes $\mathcal{Y}_S$, a the attribute vector, and t the textual description. Our goal is to generate images of unseen classes $\mathcal{Y}_U$, where $\mathcal{Y}_S \cap \mathcal{Y}_U = \emptyset$, using a generative model G that synthesizes an image $\hat{x}$ conditioned on attribute vector a and text description t: $\hat{x} = G(a, t; \theta)$.

We emphasize that our VQ-VAE, diffusion model, and all components (except the pre-trained ResNet-50 classifier backbone) are trained exclusively on seen class images. The ResNet-50 classifier is fine-tuned only on seen class images to predict attributes. No unseen class images are used during any training phase, ensuring true zero-shot evaluation. This data-efficient approach requires only thousands of samples compared to the billions used by large-scale models.

2.2. Vector Quantized Variational Autoencoder

Our VQ-VAE consists of an encoder E, decoder D, and codebook $\mathcal{Z} = \{z_k\}_{k=1}^{K}$. The encoder maps an input image to a latent representation $z_e = E(x)$, which is quantized by finding the closest codebook entry: $z_q = \arg\min_{z_k \in \mathcal{Z}} ||z_e - z_k||_2$. The decoder then reconstructs the image: $\hat{x} = D(z_q)$.

We train the VQ-VAE with a combined loss function:

$$\mathcal{L}_{\text{VQ}} = ||x - \hat{x}||_2^2 + ||sg[z_e] - z_q||_2^2 + \beta ||z_e - sg[z_q]||_2^2 \quad (1)$$

where $sg[\cdot]$ is the stop-gradient operator and β weights the commitment loss. We also incorporate LPIPS perceptual loss and an adversarial loss to improve visual quality while maintaining computational efficiency.

2.3. Conditional Latent Diffusion Model

Using latent representations from the VQ-VAE, we train a conditional diffusion model. The forward process adds Gaussian noise over T timesteps with transition probability:

$$q(z_t | z_{t-1}) = \mathcal{N}(z_t; \sqrt{1 - \beta_t} z_{t-1}, \beta_t \mathbf{I}) \quad (2)$$

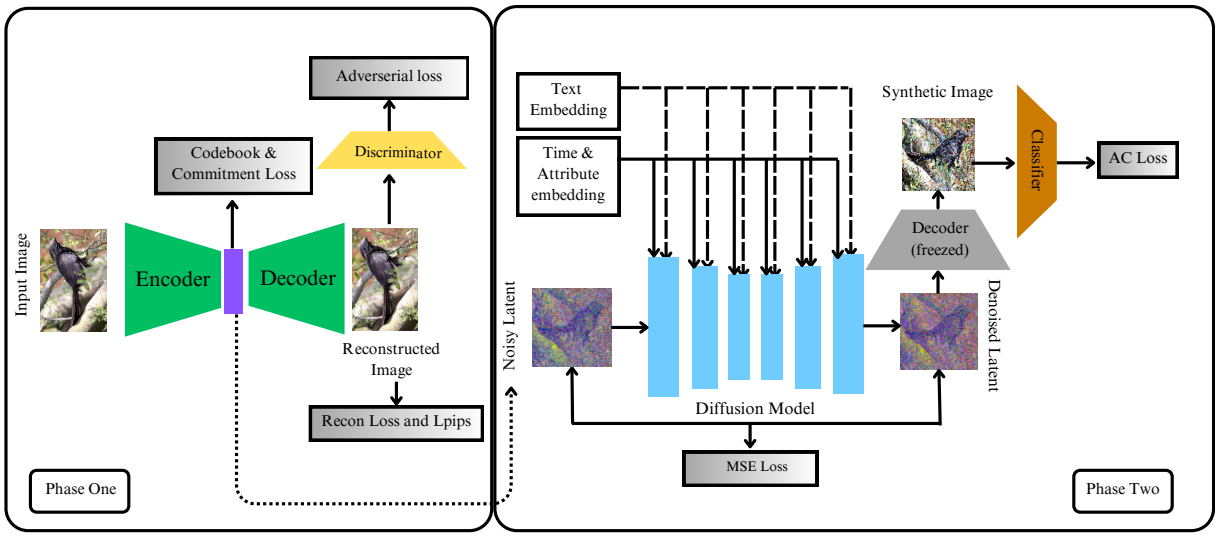

Fig. 2. Overview of the proposed AGLDM framework. Phase One trains the VQ-VAE, where high-dimensional images are mapped to a discrete latent space via a learned visual feature codebook. The encoder transforms input images into continuous latent representations, which are quantized by matching to the nearest codebook entry. The VQ-VAE is optimized using reconstruction loss, perceptual loss (LPIPS), codebook and commitment losses, along with adversarial loss from a discriminator. Phase Two trains the Latent Diffusion Model, where the diffusion process iteratively denoises latent representations while leveraging multiple conditioning mechanisms, including text embeddings, timestep embeddings, and attribute guidance. Training optimizes both the standard diffusion loss (MSE) and attribute consistency (AC) loss using a frozen classifier to ensure high-fidelity, diverse, and attribute-consistent image synthesis.

The reverse process is modeled by a neural network ϵ_θ that predicts the noise added at each timestep, conditioned on the noisy latent z_t, timestep t, and conditioning information c:

$$\mathcal{L}_{\text{LDM}} = E_{z_0,\epsilon,t,c}[||\epsilon - \epsilon_\theta(z_t, t, c)||_2^2] \quad (3)$$

Our diffusion model architecture uses a U-Net with multi-head attention and cross-attention layers, integrating conditioning through:

Text Conditioning: Text embeddings are processed through cross-attention layers at multiple levels of the U-Net, allowing the model to attend to different parts of the text description when generating different image regions.

Attribute Conditioning: Attribute vectors $a \in \{0,1\}^d$ are embedded and added to the timestep embedding: $\hat{e}_t = e_t + e_a$, ensuring attribute information flows through all network layers.

Attribute-based Classifier Guidance: We train a ResNet-50 classifier to predict attribute vectors from images, defining a self-consistency loss:

$$\mathcal{L}_{\text{self}} = -\frac{1}{D}\sum_{i=1}^{D}\Big[a_i \log(\hat{a}_{t,i}) + (1-a_i)\log(1-\hat{a}_{t,i})\Big] \quad (4)$$

The total loss becomes $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{LDM}} + \lambda\mathcal{L}_{\text{self}}$, creating a closed-loop system that ensures semantic alignment while maintaining computational efficiency.

2.4. Training and Inference

During training, we add noise to real image latent representations, train the model to predict this noise, and use the attribute classifier to ensure consistency between generated images and conditioning attributes. This efficient training process requires only dataset-specific samples rather than massive web-scale datasets, making our approach accessible to researchers with standard computational resources.

3. EXPERIMENTS & RESULTS

In this section, we describe our experimental setup, including dataset details, evaluation metrics, and comparison methods. We then present our results on zero-shot image generation tasks, emphasizing the data efficiency and computational accessibility of our approach.

3.1. Datasets & Evaluation Metrics

We evaluate our method on three standard zero-shot learning benchmarks: CUB [7], SUN [8], and AWA2 [9].

CUB contains 11,788 images of 200 bird species with 312 binary attributes and textual descriptions. SUN includes 14,340 images across 717 scene categories, with textual descriptions generated using BLIP. AWA2 consists

Table 1. Quantitative evaluation on unseen classes of the CUB, SUN, and AWA2 datasets under the ZSL setting. Our AGLDM achieves competitive performance using orders of magnitude fewer training samples. DALL-E and LAFITE were not evaluated on SUN and AWA2.

Dataset	Method	FID↓	IS↑	CLIP↑
CUB	DALL-E	56.00	2.60	–
	LAFITE	**27.53**	4.32	–
	SDXL	28.21	3.88	**0.35**
	AGLDM (Ours)	32.45	**4.57**	0.32
SUN	SDXL	**23.24**	4.23	**0.36**
	AGLDM (Ours)	45.63	**5.34**	0.33
AWA2	SDXL	**24.54**	**5.56**	**0.36**
	AGLDM (Ours)	48.72	5.21	0.33

of 37,322 images of 50 animal classes with 85 binary attributes per class, also using BLIP-generated descriptions. For all datasets, we follow the standard splits proposed by Xian et al.

All models are evaluated exclusively on unseen classes that were completely absent from training data, ensuring true zero-shot evaluation. Importantly, our experiments were conducted using a single NVIDIA RTX A5000 GPU with training data consisting of only thousands of samples per dataset (8,855 for CUB, 10,320 for SUN, 23,527 for AWA2), demonstrating the accessibility of our approach compared to large-scale models requiring billions of samples and industrial-scale computational infrastructure.

We evaluate our method using three standard metrics: Fréchet Inception Distance (FID) [10], which measures distribution similarity between real and generated images (lower is better); Inception Score (IS) [11], which evaluates image quality and diversity (higher is better); and CLIP Score, which assesses semantic alignment between generated images and their corresponding text descriptions (higher is better).

3.2. Comparative Results

We compare our approach with three state-of-the-art text-to-image models: DALL-E [3], LAFITE [12], and Stable Diffusion XL (SDXL) [13]. Table 1 presents quantitative comparison across CUB, SUN, and AWA2 datasets.

DALL-E, LAFITE, and SDXL are trained on massive datasets with enormous computational resources. DALL-E uses over 12 billion parameters trained on 250 million image-text pairs, while SDXL uses billions of image-text pairs and massive computational clusters.

SDXL achieves strong FID scores on SUN (23.24) and AWA2 (24.54), leveraging extensive pretraining on billions of samples. However, our approach achieves competitive results despite being trained exclusively on individual datasets with only thousands of samples, representing a 4-5 orders of magnitude reduction in training data requirements.

AGLDM demonstrates state-of-the-art Inception Score for CUB (**4.57**) and SUN (**5.34**), outperforming all comparison methods including SDXL. This indicates our attribute-guided approach effectively generates varied, high-quality images of unseen classes. While SDXL achieves higher CLIP scores, our model maintains competitive performance (0.33) despite using orders of magnitude less training data.

These results demonstrate that our attribute-guided approach enables effective knowledge transfer from seen to unseen classes without requiring massive datasets and computational resources. Our domain-specific approach achieves superior performance in image diversity (IS) across CUB and SUN datasets, highlighting the effectiveness of our method for accessible zero-shot image generation with dramatically reduced computational requirements.

The generated images in Figure 3 demonstrate high visual quality and accurately capture distinctive characteristics of each unseen class across all three datasets. For CUB (first row), AGLDM successfully reproduces fine-grained details such as bill shapes, wing patterns, and color distributions while maintaining natural bird poses. For AWA2 (second row), AGLDM renders diverse animal species with appropriate anatomical features, textures, and poses, capturing distinctive characteristics such as fur patterns, body shapes, and habitat contexts. For SUN (third row), our model captures essential visual elements and spatial arrangements characteristic of diverse scene environments with coherent structural composition and appropriate lighting conditions. The model's ability to generate such realistic and class-specific images without having seen examples of these classes during training highlights the effectiveness of our data-efficient text-and-attribute guided approach.

3.3. Ablation Studies

To understand the impact of different conditioning mechanisms, we conducted ablation studies comparing our full AGLDM model with three variants across all datasets. As shown in Table 2, we evaluated each variant using FID, Inception Score (IS), and CLIP Score metrics.

As evident from Table 2, text-only conditioning (AGLDM-wt) consistently outperforms attribute-only conditioning (AGLDM-wa) across all datasets, suggesting that textual descriptions provide richer semantic information than binar y attributes alone. For CUB and SUN datasets, our full AGLDM model achieves the best performance across all metrics. For AWA2, while our full model achieves the best FID (38.72) and CLIP Score (0.35), it shows a slightly lower IS (4.21) compared to the variant without classifier guidance, suggesting domain-specific trade-offs.

The results confirm that combining both textual and attribute conditioning with classifier guidance yields superior

Fig. 3. Sample synthetic images from unseen classes generated by the proposed AGLDM model. The text prompt used to generate each image is mentioned below the image.

Table 2. Ablation study: wa = without attribute, wt = without text, w/o cls = without classifier guidance.

Dataset	Method	FID↓	IS↑	CLIP↑
CUB	AGLDM-wa	68.86	3.37	0.25
	AGLDM-wt	57.42	3.23	0.28
	AGLDM w/o cls	34.45	4.56	0.30
	AGLDM	**32.45**	**4.57**	**0.32**
SUN	AGLDM-wa	74.23	3.12	0.22
	AGLDM-wt	62.78	3.51	0.27
	AGLDM w/o cls	51.72	4.14	0.31
	AGLDM	**45.63**	**5.14**	**0.33**
AWA2	AGLDM-wa	52.43	4.87	0.28
	AGLDM-wt	47.65	5.12	0.31
	AGLDM w/o cls	40.17	**5.05**	0.32
	AGLDM	**38.72**	4.21	**0.35**

semantic alignment in zero-shot image generation, while maintaining the data efficiency that makes our approach accessible to researchers with limited computational resources.

4. CONCLUSION

In this paper, we introduced AGLDM, a novel data-efficient latent diffusion framework for zero-shot text-to-image synthesis guided by object attributes. Our approach effectively generates high-quality, semantically meaningful images of previously unseen objects by leveraging the complementary strengths of textual descriptions and attribute vectors. Most importantly, we demonstrate that competitive zero-shot text-to-image generation can be achieved using only thousands of training samples on a single NVIDIA RTX A5000 GPU, representing a significant order of magnitude reduction in data requirements compared to existing large-scale models.

Experiments across CUB, SUN, and AWA2 datasets demonstrate that our attribute-guided approach with self-consistency loss produces superior images compared to text-only or attribute-only alternatives while maintaining exceptional data efficiency. Our results reveal domain-specific characteristics in zero-shot generation performance. For CUB, our model achieves optimal sample quality and diversity with competitive distribution matching. For SUN and AWA2, we establish competitive benchmarks for zero-shot text-to-image generation in these domains, utilizing resource-constrained settings.

Our work demonstrates that effective zero-shot text-to-image generation can be achieved using domain-specific training data with relatively low computational resources, making this technology accessible for specialized applications without requiring industrial-scale computational resources. This accessibility opens new opportunities for researchers and practitioners in domains with limited visual examples, such as rare species identification, medical imag-

ing, and specialized industrial applications.

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